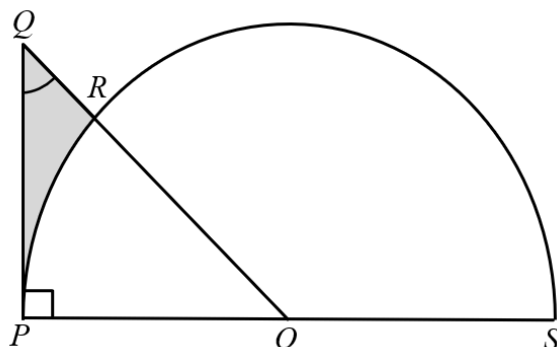


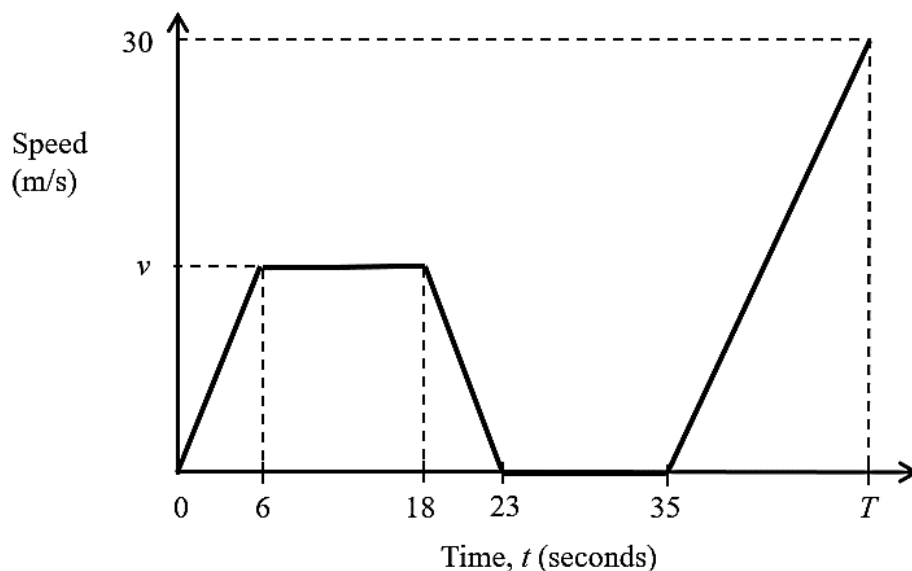
- 18 The diagram shows a semicircle with centre O and radius 8 cm.
 OP is perpendicular to PQ .
Angle $PQO = 0.7$ radians.



Find the area of the shaded region.

Answer _____ cm^2 [4]

- 19 The diagram below shows the speed-time graph for a car travelling on the road.



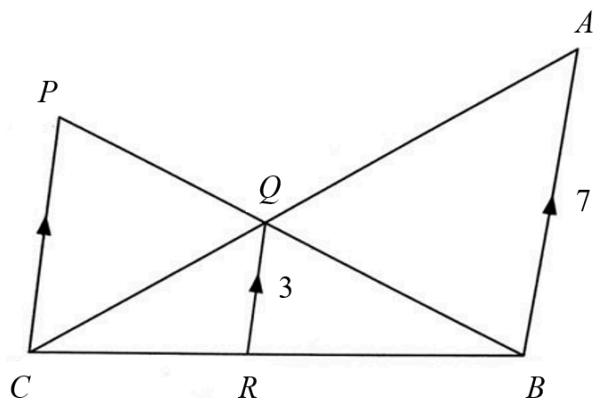
- (a) The total distance travelled by the car from 0 to 23 seconds is 385 metres. Calculate the value of v .

Answer $v =$ _____ [2]

- (b) When $t = 45$, the car is travelling at a speed of 25 m/s.
When $t = T$, the car is travelling at a speed of 30 m/s.
Find the value of T .

Answer $T =$ _____ [2]

- 20** In the diagram, AB , QR and PC are parallel.
 PQB , CQA and CRB are straight lines.
 $AB = 7$ cm and $QR = 3$ cm.



- (a) Show that triangle PCB and triangle QRB are similar.
 Give a reason for each statement.

Answer

[2]

- (b) Write down another pair of similar triangles.

Answer Triangles _____ and _____ [1]

- (c) Calculate PC .